

Alt Mobility – Data Analysis Summary Report

(All SQL queries were written and successfully executed using an online SQLite platform. For convenience and clarity, the final versions of these queries have been documented and organized in a notebook file.)

- Overview

This report summarizes the key findings from the data analysis assignment for Alt Mobility. The analysis was performed using SQL and Power BI on two datasets: Copy of customer_orders and Copy of payments.

-- Key Insights

1. Order and Sales Analysis

- Monthly sales trends show fluctuations in order volume and revenue.
- Highest revenue was consistently from orders marked as "Delivered".
- Cancellations formed a small percentage, indicating strong fulfillment.

2. Customer Behavior

- A significant portion of customers placed repeat orders.
- Most repeat orders occurred within 1–2 months of the first purchase.
- A small subset of loyal customers contributes to a large portion of revenue.

3. Payment Status Analysis

- Over 95% of payments were marked as "Success".
- Failed payments occurred mostly with certain payment methods (can be optimized).

4. Order Details Report

- A joined view of customer orders and payments was created for a full transactional log.
- Key columns like order status, payment method, and shipping address were retained.

- Visual Insights – Customer Retention (Power BI)

- A Matrix visual tracks monthly cohorts and their retention in following months.
- Cohorts show a common drop-off trend after 1 month.

- Tools Used

- SQLite: For querying and transforming data.
- Power BI: For visualizations and dashboard creation.